Multi-Agent Memory Blueprint v1

Core Goal

Build a scalable multi-agent memory architecture for OpenClaw using Supermemory as the memory/context layer, Paperclip as the organizational/orchestration layer, and OpenClaw as the runtime/session layer.

Platform Roles

- **OpenClaw**: agent runtime, messaging, sessions
- **Paperclip**: organizational structure, governance, tasks, companies
- **Supermemory**: layered memory and recall

Memory Hierarchy

0. Global Governance

Purpose: System-wide rules, governance, coordination laws, permanent standards.

Access: - read: operatives and orchestrator only - write: orchestrator only

Examples: - handoff protocol - escalation rules - write/promotion policy - permanent coordination standards

Container: `gov_global`

1. Company Memory

Purpose: Durable company/domain knowledge shared across projects in that company.

Examples: - website company stack standards - game company rendering pipeline - preferred tools - reusable workflows - domain-specific methods

Container examples: - `company_web` - `company_games`

2. Project Shared Memory

Purpose: Live collaboration memory for active projects.

Examples: - blockers - decisions - active architecture choices - handoff status - project constraints - verified findings

Container examples: - `project_supermemory_fork` - `project_games_prototype_alpha`

3. Restricted Shared Memory

Purpose: Shared memory for limited groups without exposing full company scope.

Examples: - `restricted_billing` - `restricted_admin` - `restricted_render_ops`

4. Agent Private Memory

Purpose: Local durable working memory for the individual agent.

Examples: - local heuristics - unfinished patterns - personal workflow preferences - non-shared useful notes

Container examples: - `agent_orchestrator_private` - `agent_014_private`

5. Agent Experience

Purpose: Curated long-term specialization knowledge.

Storage form: Markdown files, primarily `experience.md` files.

Examples: - repeated bottlenecks - durable lessons - successful playbooks - known failure modes

6. User Memory

Purpose: User-specific persistent preferences and facts.

Container example: `user_<id>`

7. Session Memory

Purpose: Active thread/session continuity.

Mechanism: `conversationId`

8. Ephemeral

Purpose: Temporary context that should usually never become durable memory.

9. Cold Storage

Purpose: Archived completed project memory, isolated from normal agents.

Access: - orchestrator - operators/managers on command

Not for routine agent recall.

Container examples: - `archive_project_supermemory_fork` - `archive_project_web_redesign`

Access Model

Orchestrator

Can: - create memory spaces - grant/revoke access - inspect private memory if needed - promote project learnings upward - archive finished projects - retrieve cold storage on command

Operators / Managers

Can: - access assigned company/project memories - access cold storage when authorized - help curate durable knowledge - read global governance if designated as operatives

Agents

Can: - use private memory - use assigned project memory - use only explicitly granted company/restricted memory - not access cold storage by default - not create new shared memory spaces

Write Policy

Direct writes allowed

- own private memory
- assigned project memory if useful and non-noisy

Proposed writes

- company memory
- restricted shared memory

Orchestrator-only writes

- governance memory
- creation of new company/project/restricted/cold-storage spaces
- archival metadata

Promotion Flow

- private -> project -> company -> governance

Promote upward only when value is proven and reusable.

Project Completion Lifecycle

1. Project completes.
2. Distill bottlenecks, issues, and durable lessons.
3. Write curated lessons into the specialized agent's `experience.md`.
4. Wipe most specialized agent project memory after completion.
5. Archive project memory to cold storage.
6. Revoke routine agent access to archived project memory.
7. Carry only curated `experience.md` knowledge forward to future projects.

Metadata Schema for Supermemory

```
{
  "agent_id": "agent_orchestrator",
  "agent_name": "Orchestrator",
  "agent_kind": "orchestrator",
  "company_id": "company_web",
  "project_id": "project_supermemory_fork",
  "memory_scope": "project",
  "visibility": "shared",
  "written_by": "agent_orchestrator",
  "source_session": "discord_1485839353149391020",
  "channel": "discord",
  "confidence": 0.87,
  "importance": "high",
  "retention": "project_lifecycle",
  "status": "active",
  "tags": ["architecture", "memory-policy", "handoff"],
  "approved_by": "agent_orchestrator",
  "promoted_from": "agent_private",
  "timestamp": "2026-03-24T03:15:00Z"
}
```

Suggested Enum Values

`memory_scope`

- `governance`
- `company`
- `project`
- `restricted_shared`
- `agent_private`
- `experience`
- `user`
- `session`
- `cold_storage`
- `ephemeral`

visibility

- shared
- restricted
- private
- orchestrator_only
- operator_only
- manager_operator_only

retention

- ephemeral
- short_term
- project_lifecycle
- persistent
- archival

status

- proposed
- approved
- active
- archived
- stale
- rejected

Naming Conventions

- gov_global
- company_web
- company_games
- project_supermemory_fork
- restricted_billing
- restricted_admin
- agent_orchestrator_private
- agent_001_private
- archive_project_supermemory_fork
- user_<id>

Use `conversationId` for thread/session context.

Workspace File Tree

```
agents/
  BLUEPRINT.md
  SHARED_PLAYBOOK.md
  MEMORY_POLICY.md
  ORCHESTRATOR.md

  registry/
```

```
    agents.json
    companies.json
    projects.json
    memory_spaces.json

  companies/
    _template.company.md
    web.md
    games.md

  projects/
    _template.project.md
    supermemory-fork.md

  agent-profiles/
    _template.agent.md
    orchestrator.md

  experience/
    orchestrator.md
    _template.experience.md
```

Templates

Company Template

```
# Company: <company_id>

## Mission
<what this company specializes in>

## Durable Knowledge
- stack/tooling
- standards
- workflows
- domain patterns

## What Belongs In Company Memory
- reusable methods
- domain standards
- stable tool preferences

## What Stays Out
- raw project chatter
- private agent notes
- one-off noise

## Access
- who can read
```

```
- who can write
- who can approve promotions
```

Project Template

```
# Project: <project_id>

## Objective
<goal>

## Company
<company_id>

## Assigned Agents
- <agent_ids>

## Allowed Memory
- project shared memory
- relevant company memory
- agent private memory

## Deliverables
- <deliverables>

## Completion Rules
- archive to cold storage
- extract valuable lessons to specialist experience.md
- wipe most specialized project memory after completion
- revoke unnecessary access
```

Agent Profile Template

```
# Agent Profile: <agent_id>

## Name
<display name>

## Mission
<current purpose>

## Tone
<tone>

## Permissions
- tools:
- memory reads:
- memory writes:
- escalation rights:

## Memory Behavior
```

```
- what to write
- what to avoid
- when to escalate instead of writing

## Handoff Style
<how this agent communicates state>

## Constraints
<what this agent must never do>
```

Experience Template

```
# Experience: <agent_id or specialization>

## Durable Lessons
- lesson:
  context:
  signal:
  action:

## Bottlenecks Seen Repeatedly
- <pattern>

## What Works
- <best practices>

## What Fails
- <anti-patterns>

## Promotion Candidates
- lessons that may deserve company-level memory
```

Registry Examples

agents.json

```
[
  {
    "agent_id": "agent_orchestrator",
    "display_name": "Orchestrator",
    "status": "active",
    "capabilities": ["coordination", "routing", "memory-
governance"],
    "assigned_projects": ["project_supermemory_fork"],
    "private_memory_id": "agent_orchestrator_private",
    "experience_file": "agents/experience/orchestrator.md",
    "memory_permissions": {
      "read": ["gov_global", "company_web",
"project_supermemory_fork"],
```

```
              "write": ["agent_orchestrator_private",
"project_supermemory_fork", "gov_global"],
              "admin": true
            }
          }
        ]
```

companies.json

```
        [
          {
            "company_id": "company_web",
            "name": "Web Company",
            "status": "active",
            "memory_id": "company_web",
            "projects": ["project_supermemory_fork"],
            "description": "Website and software delivery company"
          },
          {
            "company_id": "company_games",
            "name": "Games Company",
            "status": "planned",
            "memory_id": "company_games",
            "projects": [],
            "description": "Game, art, rendering, and asset pipeline
company"
          }
        ]
```

projects.json

```
        [
          {
            "project_id": "project_supermemory_fork",
            "company_id": "company_web",
            "status": "active",
            "memory_id": "project_supermemory_fork",
            "archive_memory_id": "archive_project_supermemory_fork",
            "assigned_agents": ["agent_orchestrator"],
            "completion_policy": {
              "archive_on_close": true,
              "extract_experience": true,
              "wipe_specialized_memory": true,
              "cold_storage_access": ["agent_orchestrator",
"operators", "managers"]
            }
          }
        ]
```

memory_spaces.json

```
[
  {
    "memory_id": "gov_global",
    "scope": "governance",
    "status": "active",
    "readers": ["agent_orchestrator", "operatives"],
    "writers": ["agent_orchestrator"],
    "retention": "persistent"
  },
  {
    "memory_id": "company_web",
    "scope": "company",
    "status": "active",
    "readers": ["agent_orchestrator"],
    "writers": ["agent_orchestrator"],
    "retention": "persistent"
  },
  {
    "memory_id": "project_supermemory_fork",
    "scope": "project",
    "status": "active",
    "readers": ["agent_orchestrator"],
    "writers": ["agent_orchestrator"],
    "retention": "project_lifecycle"
  },
  {
    "memory_id": "archive_project_supermemory_fork",
    "scope": "cold_storage",
    "status": "inactive",
    "readers": ["agent_orchestrator", "operators",
"managers"],
    "writers": ["agent_orchestrator"],
    "retention": "archival"
  }
]
```